Akshat Krishan

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SUMMARY

Full-Stack & Cloud-Native Engineer specializing in TypeScript, Next.js, NestJS, and scalable backend infrastructure. Experienced in designing multi-database architectures (PostgreSQL, DynamoDB, MongoDB, Redis), distributed background processing pipelines with BullMQ and Redlock, and managing production Kubernetes (GKE) deployments via automated GitOps pipelines with Helm and ArgoCD.

EXPERIENCE

Full-Stack Engineer (Freelance), Remote — Dec 2025 – Jan 2026

- Designed and normalized PostgreSQL schemas on Supabase for a job board platform, implementing composite indexes and foreign-key optimizations that reduced read query latency by 25%.
- Engineered an end-to-end authentication and checkout workflow integrating Clerk OTP verification and Razorpay webhooks, decreasing user onboarding and checkout drop-off by 40%.
- Built a modular, multi-step candidate application flow using React Hook Form, Zod, and Zustand, isolating input state to eliminate redundant component re-renders and improving form completion by 30%.

Software Engineer Intern, Intern Labs, Remote — July 2025 – Sept 2025

- Architected a full-stack Turborepo monorepo with Next.js, consolidating shared UI components, shared TypeScript configs, and NextAuth role-based access control (RBAC) across separate Admin and Candidate portals.
- Integrated an isolated code execution engine on AWS EC2 using Judge0 and Supabase (PostgreSQL) to evaluate custom programming submissions in real time, sustaining 99.9% system uptime.
- Optimized client-side data fetching and cache invalidation policies with SWR, improving admin dashboard load speeds by 40% during live candidate assessments and scoring rounds.

PROJECTS

Ascend — GitHub | System Architecture (C4)

- **Data Architecture & Storage:** Partitioned workloads across databases by access pattern: PostgreSQL for relational entities, DynamoDB for append-only audit logging, and MongoDB with 30-day TTL indexes to prevent Postgres table bloat; encoded monthly habit activity into 31-byte bitmasks for $O(1)$ single-row heatmap reads.
- **Distributed Tasks & Concurrency:** Prevented full-table scans by bucketing user resets into hourly BullMQ jobs grouped by IANA timezone, pairing with Upstash Redis and Redlock distributed mutexes to batch-flush user progress to PostgreSQL every 10 seconds without write race conditions.
- **Modular Monolith on GKE:** Containerized and deployed a modular NestJS monolith alongside Next.js on Google Kubernetes Engine (GKE), configuring Horizontal Pod Autoscaling (3–8 pods), GCE Ingress TLS, multi-pod Socket.IO sync via Redis Pub/Sub, and an in-cluster Redis instance backed by a GCP Persistent Volume Claim (PVC) for persistent queues.
- **GitOps & Automated Releases:** Configured automated zero-downtime deployments using Helm charts and ArgoCD, managed encrypted secrets with Bitnami SealedSecrets, ran automated migration jobs via Helm hooks, and secured CI/CD with GCP OIDC Workload Identity Federation.

EDUCATION

Rajasthan Technical University, Kota — 2022 – 2026
B.Tech in Computer Science and Engineering — **CGPA: 8.45/10**

TECHNICAL SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), SQL, Bash, YAML, HTML5, CSS3
- **Backend & Architecture:** NestJS (Modular Architecture), Node.js, Express.js, BullMQ, Redlock, Socket.IO (Redis Pub/Sub), REST APIs, JWT, OAuth 2.0, Web Push
- **Databases & Caching:** PostgreSQL, Prisma ORM, AWS DynamoDB (Single-Table Design), MongoDB, Redis (Upstash, Caching, AOF Persistence), Supabase
- **Cloud, DevOps & Tooling:** Google Cloud Platform (GCP / GKE, Persistent Disks), AWS (EC2, S3, DynamoDB), Kubernetes (HPA, PVC, Helm), ArgoCD (GitOps), Bitnami SealedSecrets, Docker, GitHub Actions (CI/CD), Turborepo
- **Frontend & Testing:** Next.js 16 (App Router, Server Components), React.js, Tailwind CSS, Zustand, SWR, React Hook Form, Zod, Vitest (Unit, Integration)